

Schur-Complement Certificates for Light Vertex Sets: Rigorous Lemmas and Probabilistic Bounds

1 Basic definitions

Let $G = (V, E)$ be a finite, simple, undirected graph. Write $n := |V|$. Let L be the (combinatorial) Laplacian of G :

$$L_{uu} = \deg(u), \quad L_{uv} = \begin{cases} -1 & \{u, v\} \in E, \\ 0 & \text{otherwise.} \end{cases}$$

For $S \subseteq V$, let $\tilde{L}_S$ denote the Laplacian of the induced subgraph $G[S]$ viewed as a matrix on $\mathbb{R}^S$. We also view it as a matrix on $\mathbb{R}^V$ by padding with zeros on $V \setminus S$; we denote this padded matrix by L_S :

$$L_S := P_S^\top \tilde{L}_S P_S,$$

where $P_S : \mathbb{R}^V \rightarrow \mathbb{R}^S$ is the coordinate projection.

Definition 1 (ε -light set). *Fix $\varepsilon \in (0, 1)$. A set $S \subseteq V$ is called ε -light if*

$$\varepsilon L - L_S \succeq 0$$

as a matrix on $\mathbb{R}^V$.

2 Schur complement characterization

Fix a partition $V = S \sqcup \bar{S}$ and block-decompose

$$L = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix},$$

where A is indexed by S and C by $\bar{S}$. Let $C^\dagger$ denote the Moore–Penrose pseudoinverse of C .

Definition 2 (Schur-complement Laplacian). *Define the (generalized) Schur complement*

$$L_S^{\text{Schur}} := A - BC^\dagger B^\top,$$

a symmetric positive semidefinite matrix on $\mathbb{R}^S$ with $L_S^{\text{Schur}} \mathbf{1}_S = 0$.

Lemma 1 (Dirichlet variational identity). *For every $z \in \mathbb{R}^S$,*

$$z^\top L_S^{\text{Schur}} z = \min_{w \in \mathbb{R}^{\bar{S}}} \begin{pmatrix} z \\ w \end{pmatrix}^\top \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} \begin{pmatrix} z \\ w \end{pmatrix}.$$

Proof. Let $Q(w) := \begin{pmatrix} z \\ w \end{pmatrix}^\top L \begin{pmatrix} z \\ w \end{pmatrix} = z^\top A z + 2z^\top B w + w^\top C w$. This is a convex quadratic in w (since $C \succeq 0$). Minimizers satisfy the normal equation $Cw = -B^\top z$ in the least-squares sense; the minimum value is obtained by $w^* = -C^\dagger B^\top z$ (up to addition of $\ker(C)$), and substituting yields

$$\min_w Q(w) = z^\top A z - z^\top B C^\dagger B^\top z = z^\top (A - B C^\dagger B^\top) z.$$

□

Proposition 1 (Schur characterization of lightness). *For $S \subseteq V$ and $\varepsilon \in (0, 1)$,*

$$\varepsilon L - L_S \succeq 0 \iff \tilde{L}_S \preceq \varepsilon L_S^{\text{Schur}} \text{ on } \mathbb{R}^S.$$

Equivalently, for all $z \in \mathbb{R}^S$,

$$z^\top \tilde{L}_S z \leq \varepsilon z^\top L_S^{\text{Schur}} z.$$

Proof. ($\Rightarrow$) Assume $\varepsilon L - L_S \succeq 0$. Fix $z \in \mathbb{R}^S$ and consider any extension $y = \begin{pmatrix} z \\ w \end{pmatrix} \in \mathbb{R}^V$. Since L_S depends only on z ,

$$0 \leq y^\top (\varepsilon L - L_S) y = \varepsilon y^\top L y - z^\top \tilde{L}_S z.$$

Thus $z^\top \tilde{L}_S z \leq \varepsilon y^\top L y$ for all w , hence

$$z^\top \tilde{L}_S z \leq \varepsilon \min_w y^\top L y = \varepsilon z^\top L_S^{\text{Schur}} z$$

by Lemma 1.

($\Leftarrow$) Conversely, assume $z^\top \tilde{L}_S z \leq \varepsilon z^\top L_S^{\text{Schur}} z$ for all z . For any $y = \begin{pmatrix} z \\ w \end{pmatrix}$ we have $y^\top L y \geq \min_{w'} \begin{pmatrix} z \\ w' \end{pmatrix}^\top L \begin{pmatrix} z \\ w' \end{pmatrix} = z^\top L_S^{\text{Schur}} z$ (Lemma 1). Hence

$$y^\top (\varepsilon L - L_S) y = \varepsilon y^\top L y - z^\top \tilde{L}_S z \geq \varepsilon z^\top L_S^{\text{Schur}} z - z^\top \tilde{L}_S z \geq 0.$$

□

3 Effective cut functional and necessary conditions

Definition 3 (Effective cut functional). *For $T \subseteq S$, define*

$$\Phi_S(T) := \mathbf{1}_T^\top L_S^{\text{Schur}} \mathbf{1}_T.$$

Lemma 2 (Two-terminal Dirichlet form). *For $T \subseteq S$,*

$$\Phi_S(T) = \min\{y^\top L y : y|_T = \mathbf{1}, y|_{S \setminus T} = 0\},$$

where the minimization is over all $y \in \mathbb{R}^V$ satisfying the boundary constraints.

Proof. Apply Lemma 1 with $z = \mathbf{1}_T \in \mathbb{R}^S$. □

Proposition 2 (Effective cut-test is necessary). *If S is ε -light, then for every $T \subseteq S$,*

$$|E(T, S \setminus T)| \leq \varepsilon \Phi_S(T).$$

Proof. By Proposition 1, $\tilde{L}_S \preceq \varepsilon L_S^{\text{Schur}}$. Test this inequality on $\mathbf{1}_T \in \mathbb{R}^S$:

$$\mathbf{1}_T^\top \tilde{L}_S \mathbf{1}_T \leq \varepsilon \mathbf{1}_T^\top L_S^{\text{Schur}} \mathbf{1}_T.$$

Since $\tilde{L}_S$ is the Laplacian of $G[S]$, $\mathbf{1}_T^\top \tilde{L}_S \mathbf{1}_T = |E(T, S \setminus T)|$. □

4 A counterexample: the raw cut-test is not sufficient

The inequality in Proposition 2 involves $\Phi_S(T)$, not the raw boundary size $\delta(T) := |E(T, V \setminus T)|$. The latter is insufficient to guarantee lightness.

Proposition 3 (Raw cut-test is not sufficient). *Fix $\varepsilon \in (0, 1)$. There exist G and $S \subseteq V(G)$ such that for all $T \subseteq S$,*

$$|E(T, S \setminus T)| \leq \varepsilon |E(T, V \setminus T)|,$$

but S is not ε -light.

Proof. Let $M \geq 1/\varepsilon - 1$ be an integer. Let

$$V = \{u, v\} \sqcup A \sqcup B, \quad |A| = |B| = M,$$

with edges: $\{u, v\}$, all $\{u, a\}$ for $a \in A$, and all $\{v, b\}$ for $b \in B$. Let $S = \{u, v\}$.

For $T = \{u\}$ we have $|E(T, S \setminus T)| = 1$ and $|E(T, V \setminus T)| = \deg(u) = 1 + M$, so $1 \leq \varepsilon(1 + M)$ by the choice of M . Similarly for $T = \{v\}$; the cases $T = \emptyset, S$ are trivial.

Now define $x \in \mathbb{R}^V$ by $x_u = 1$, $x_v = 0$, $x_a = 1$ for $a \in A$, and $x_b = 0$ for $b \in B$. Then $x^\top L_S x = (x_u - x_v)^2 = 1$ while $x^\top L x = 1$ (all other edges see zero difference). Hence

$$x^\top (\varepsilon L - L_S) x = \varepsilon - 1 < 0,$$

so $\varepsilon L - L_S \not\geq 0$. □

5 A fully explicit lower bound on $\Phi_S(T)$ (two-layer bridging)

For $S \subseteq V$ and $T \subseteq S$, define for each $w \in \bar{S}$:

$$a_w := |E(w, T)|, \quad b_w := |E(w, S \setminus T)|.$$

Lemma 3 (Two-layer conductance lower bound). *For every $S \subseteq V$ and $T \subseteq S$,*

$$\Phi_S(T) \geq |E(T, S \setminus T)| + \sum_{w \in \bar{S}} \frac{a_w b_w}{a_w + b_w},$$

with the convention $\frac{ab}{a+b} = 0$ if $ab = 0$. In particular,

$$\Phi_S(T) \geq |E(T, S \setminus T)| + \frac{1}{2} \sum_{w \in \bar{S}} \min(a_w, b_w).$$

Proof. Consider the subgraph H obtained from G by *deleting* all edges except: (i) edges in $E(T, S \setminus T)$, (ii) edges from $\bar{S}$ into T , and (iii) edges from $\bar{S}$ into $S \setminus T$. Deleting edges can only *decrease* effective conductance between the terminal sets (by Rayleigh monotonicity / Dirichlet principle), hence by Lemma 2,

$$\Phi_S(T) = \min\{y^\top L_G y : y|_T = 1, y|_{S \setminus T} = 0\} \geq \min\{y^\top L_H y : y|_T = 1, y|_{S \setminus T} = 0\}.$$

Now contract all vertices of T into a single source node s and all vertices of $S \setminus T$ into a single sink node t . In the contracted network:

- each edge in $E(T, S \setminus T)$ becomes a unit-conductance edge directly between s and t , contributing conductance $|E(T, S \setminus T)|$ in parallel;
- each $w \in \bar{S}$ yields a_w parallel unit edges between s and w (conductance a_w) and b_w parallel unit edges between w and t (conductance b_w); these two bundles are in series, so the branch conductance is $c_w = \frac{a_w b_w}{a_w + b_w}$.

All these branches are in parallel, so total conductance (hence the Dirichlet minimum energy at unit voltage) is exactly $|E(T, S \setminus T)| + \sum_w c_w$. This proves the first inequality. The second follows from $\frac{ab}{a+b} \geq \frac{1}{2} \min(a, b)$. $\square$

Corollary 1 (A sufficient inequality for the effective cut-test). *If for some $T \subseteq S$ one has*

$$(1 - \varepsilon) |E(T, S \setminus T)| \leq \varepsilon \sum_{w \in \bar{S}} \frac{a_w b_w}{a_w + b_w},$$

then $|E(T, S \setminus T)| \leq \varepsilon \Phi_S(T)$.

Proof. By Lemma 3, $\Phi_S(T) \geq |E(T, S \setminus T)| + \sum_w \frac{a_w b_w}{a_w + b_w}$. The displayed inequality rearranges to $|E(T, S \setminus T)| \leq \varepsilon \Phi_S(T)$. $\square$

6 From the effective cut-test to domination on few-level vectors

In what follows, we use only that L_S^{Schur} is a Laplacian on S (symmetric PSD, zero row sums), so it can be written as the Laplacian of a weighted complete graph on S (effective conductances).

Lemma 4 (Threshold decomposition for Laplacian energies). *Let Q be any Laplacian on a finite set S . Let $x \in \mathbb{R}^S$ be nonnegative and take distinct values $0 = a_0 < a_1 < \dots < a_m$. Define threshold sets $T_i := \{v \in S : x_v \geq a_i\}$ and increments $d_i := a_i - a_{i-1}$. Then:*

$$\begin{aligned} x^\top Q x &\geq \sum_{i=1}^m d_i^2 \mathbf{1}_{T_i}^\top Q \mathbf{1}_{T_i}, \\ x^\top Q x &\leq m \sum_{i=1}^m d_i^2 \mathbf{1}_{T_i}^\top Q \mathbf{1}_{T_i}. \end{aligned}$$

Proof. Write Q as a weighted Laplacian: $x^\top Q x = \sum_{\{u,v\}} w_{uv} (x_u - x_v)^2$ with $w_{uv} \geq 0$. Fix an edge $\{u, v\}$ and assume $x_u \geq x_v$. Then

$$x_u - x_v = \sum_{i: a_i \in (x_v, x_u]} d_i,$$

so $(x_u - x_v)^2 \geq \sum_{i: a_i \in (x_v, x_u]} d_i^2$ (since $(\sum r_i)^2 \geq \sum r_i^2$ for $r_i \geq 0$), and also $(x_u - x_v)^2 \leq \#\{i : a_i \in (x_v, x_u]\} \cdot \sum d_i^2 \leq m \sum d_i^2$ (Cauchy–Schwarz). The condition $a_i \in (x_v, x_u]$ is exactly that $\mathbf{1}_{T_i}(u) \neq \mathbf{1}_{T_i}(v)$, so summing over edges gives the stated bounds. $\square$

Theorem 1 (Effective cut-test implies domination on m -level vectors). *Assume that for a set $S \subseteq V$ and some $\varepsilon \in (0, 1)$ the effective cut-test holds:*

$$\mathbf{1}_T^\top \tilde{L}_S \mathbf{1}_T \leq \varepsilon \mathbf{1}_T^\top L_S^{\text{Schur}} \mathbf{1}_T \quad \forall T \subseteq S.$$

Let $x \in \mathbb{R}^S$ be any vector that takes at most m distinct values on S . Then

$$x^\top \tilde{L}_S x \leq \varepsilon m x^\top L_S^{\text{Schur}} x.$$

Proof. By decomposing x into positive/negative parts and shifting by a constant (which does not change either Laplacian energy), it suffices to treat nonnegative x .

Let the distinct values of x be $0 = a_0 < a_1 < \dots < a_m$ and define T_i, d_i as in Lemma 4. Apply the *upper* bound of Lemma 4 to $Q = \tilde{L}_S$:

$$x^\top \tilde{L}_S x \leq m \sum_{i=1}^m d_i^2 \mathbf{1}_{T_i}^\top \tilde{L}_S \mathbf{1}_{T_i}.$$

Apply the effective cut-test to each T_i :

$$\mathbf{1}_{T_i}^\top \tilde{L}_S \mathbf{1}_{T_i} \leq \varepsilon \mathbf{1}_{T_i}^\top L_S^{\text{Schur}} \mathbf{1}_{T_i},$$

so

$$x^\top \tilde{L}_S x \leq \varepsilon m \sum_{i=1}^m d_i^2 \mathbf{1}_{T_i}^\top L_S^{\text{Schur}} \mathbf{1}_{T_i}.$$

Finally apply the *lower* bound of Lemma 4 to $Q = L_S^{\text{Schur}}$:

$$\sum_{i=1}^m d_i^2 \mathbf{1}_{T_i}^\top L_S^{\text{Schur}} \mathbf{1}_{T_i} \leq x^\top L_S^{\text{Schur}} x,$$

which completes the proof. $\square$

7 A fixed- T probabilistic certificate via bridging witnesses

This section gives a rigorous high-probability *fixed- T* criterion that implies the effective cut-test at T for a random sampled set S . It does *not* by itself guarantee the test simultaneously for all $T \subseteq S$.

Lemma 5 (Chernoff bounds). *If $X \sim \text{Bin}(N, p)$ with $\mu = pN$, then for $\eta \in (0, 1)$,*

$$\Pr[X \geq (1 + \eta)\mu] \leq \exp\left(-\frac{\eta^2 \mu}{3}\right), \quad \Pr[X \leq (1 - \eta)\mu] \leq \exp\left(-\frac{\eta^2 \mu}{2}\right).$$

Fix $T \subseteq V$. Sample each vertex into S independently with probability $p \in (0, 1)$, and condition on $T \subseteq S$. Define $\delta(T) := |E(T, V \setminus T)|$ and

$$X := |E(T, S \setminus T)|.$$

Then $X \mid (T \subseteq S) \sim \text{Bin}(\delta(T), p)$, hence $\mathbb{E}[X \mid T \subseteq S] = p\delta(T)$.

Now define the witness set

$$W(T) := \{w \in V \setminus T : |E(w, T)| \geq 1\}.$$

For $w \in W(T)$, let $A_w := N(w) \cap (V \setminus T)$ (neighbors of w outside T). Let $U := V \setminus T$ and define the overlap parameter

$$\Delta_U(T) := \max_{u \in U} |\{w \in W(T) : u \in A_w\}|.$$

Lemma 6 (Matching-based witness thinning). *There exists a subset $W' \subseteq W(T)$ and an injective map $w \mapsto u(w) \in A_w$ such that all $u(w)$ are distinct and*

$$|W'| \geq \frac{|W(T)|}{\Delta_U(T)}.$$

Proof. Construct a bipartite graph between $W(T)$ and U with edges $w \sim u$ iff $u \in A_w$. Run the greedy matching algorithm: while unmatched vertices remain in $W(T)$, pick any unmatched w and match it to any neighbor $u \in A_w$, then delete u and all its incident edges. Each time we delete a vertex $u \in U$ we remove at most $\Delta_U(T)$ vertices from $W(T)$ that could have matched to u . Therefore, after producing M matches we have removed at most $M\Delta_U(T)$ vertices from $W(T)$, so $M \geq |W(T)|/\Delta_U(T)$. $\square$

For $w \in W'$, define the independent surrogate indicator

$$J_w := \mathbf{1}_{\{w \notin S\}} \cdot \mathbf{1}_{\{u(w) \in S\}}, \quad \tilde{Z} := \sum_{w \in W'} J_w.$$

Conditional on $T \subseteq S$, the variables $\{J_w\}_{w \in W'}$ are mutually independent and satisfy $\mathbb{E}[J_w \mid T \subseteq S] = p(1-p)$, hence

$$\mu_{\tilde{Z}} := \mathbb{E}[\tilde{Z} \mid T \subseteq S] = |W'|p(1-p).$$

Lemma 7 (Fixed- T high-probability bridging domination). *Fix $\varepsilon \in (0, 1)$ and $\eta \in (0, 1)$. Conditioned on $T \subseteq S$, the event*

$$(1 - \varepsilon)X \leq \varepsilon \cdot \frac{\tilde{Z}}{2}$$

holds with probability at least

$$1 - \exp\left(-\frac{\eta^2 p \delta(T)}{3}\right) - \exp\left(-\frac{\eta^2 |W'| p(1-p)}{2}\right),$$

provided the deterministic inequality

$$(1 - \varepsilon)(1 + \eta)\delta(T) \leq \varepsilon \cdot \frac{1 - \eta}{2} |W'| (1 - p)$$

holds. In particular, using Lemma 6, it suffices that

$$|W(T)| \geq \frac{2(1 - \varepsilon)(1 + \eta)}{\varepsilon(1 - \eta)} \cdot \frac{\delta(T) \Delta_U(T)}{1 - p}.$$

Proof. By Lemma 5, with conditional probability at least $1 - \exp(-\eta^2 p \delta(T)/3)$ we have $X \leq (1 + \eta)p\delta(T)$. Also $\tilde{Z} \sim \text{Bin}(|W'|, p(1-p))$ conditional on $T \subseteq S$, so by Lemma 5, with conditional probability at least $1 - \exp(-\eta^2 |W'| p(1-p)/2)$ we have $\tilde{Z} \geq (1 - \eta)|W'|p(1-p)$. On the intersection of these two events,

$$(1 - \varepsilon)X \leq (1 - \varepsilon)(1 + \eta)p\delta(T) \leq \varepsilon \cdot \frac{1 - \eta}{2} |W'| p(1 - p) \leq \varepsilon \cdot \frac{\tilde{Z}}{2},$$

where the middle inequality is the deterministic condition and the last inequality is $\tilde{Z} \geq (1 - \eta)|W'|p(1-p)$. Finally use $|W'| \geq |W(T)|/\Delta_U(T)$. $\square$

Remark 1 (Connection to the effective cut-test). *Since $\frac{a_w b_w}{a_w + b_w} \geq \frac{1}{2} \mathbf{1}_{\{a_w \geq 1\}} \mathbf{1}_{\{b_w \geq 1\}}$ and $J_w \leq \mathbf{1}_{\{a_w \geq 1\}} \mathbf{1}_{\{b_w \geq 1\}} \mathbf{1}_{\{w \notin S\}}$, the event in Lemma 7 implies the sufficient inequality in Corollary 1 for that fixed T (up to using the surrogate $\tilde{Z}$). Thus it yields a rigorous fixed- T probabilistic certificate toward $|E(T, S \setminus T)| \leq \varepsilon \Phi_S(T)$.*